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# Label noise tolerance of deep semantic segmentation networks for extracting buildings in ultra-high-resolution aerial images of semi-built environments

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**ABSTRACT**

Freely available building maps of rapidly changing built and semi-built environments may contain label noise. When temporal correspondence between images and labels does not hold, the labels may be subject to incorrectly observed building instances. For example, in most growing semi-built environments, such as the Kutupalong mega-camp in Bangladesh, labels corresponding to a past date may not be updated or might not have been properly labelled, resulting in label noise. Tagging/labelling can be done either manually (by humans) or automatically (by a machine/model). We manually label images for our stricter evaluation regime, but a trained model can automatically label images without human supervision. Our best performing model generates labels which improve F1-score by 17.2% and improve Intersection-over-Union score by 23.2%, when compared to the fidelity of commonly used noisy labels. Our stricter evaluation regime reveals interesting insights about the paradoxical behaviour of deep neural networks in conjunction to label noise.

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Label noise; building extraction; semantic segmentation; deep learning; Rohingya

## 1. Introduction

Ever since the large-scale migration of Rohingya migrants into Bangladesh began in late 2017, the population has grown to over a million refugees, and therefore has been an increasing need for housing and aid. Refugee management at such scales can be difficult to efficiently execute and may benefit from automated methods for aiding specific tasks. For example, data-driven automated systems for refugee and disaster management of Rohingya refugee camps have been developed (Alam et al. 2020; Ahmed et al. 2020a, 2020b, 2020c; Adnan et al. 2020). The growing refugee population and along with it, the increasing number of refugee camp buildings are becoming difficult to accommodate; especially since the land is already a scarce resource in Bangladesh (one of the most densely populated countries in the world). The refugee camps themselves have evolved into large urban areas with hundreds of thousands of camp buildings, and require urban management practices similar to those required for actual large cities. The largest camp, Kutupalong mega-camp, currently houses over 600,00 refugees and has over 100,000

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camp buildings. Remotely sensed ultra-high-resolution aerial images of refugee camps collected by the United Nations International Organization for Migration using unmanned aerial vehicles with mounted cameras have been utilized for deep learning-based extraction of camp buildings (Ahmed et al. 2020b). Building maps can aid a variety of refugee management tasks such as aid dispersion and disaster management.

In most building extraction tasks, the labels are derived from freely available vector data such as those from OpenStreetMap. The presence of label noise in these data is very common, especially since these annotated data are not created while keeping applications like building extraction in mind. Volunteers performing annotation have to prioritize speed over accuracy. Furthermore, the use of labelling tools itself can be a source of label noise (Frank et al. 2017). Most building labelling tools create polygons with internal angles of 90 degrees, other non-rectangular shapes are not encouraged, as they take longer to annotate. The presence of label noise can increase the complexity of learning tasks (Garcia et al. 2015) as well as reduce classifier performance (Pelletier et al. 2017). We have already observed massive-label noise in these freely available data (Figure 1). Deep learning-based semantic segmentation models can effectively learn to produce labels which are more accurate and less noisy than the noisy distribution they have learned from (Ahmed et al. 2020b). This paradoxical behaviour of deep learning systems is a common phenomenon (Rolnick et al. 2017). This is very true for our case since most label noise occurs at random and thus there are no generalize-able patterns in the noise that the networks can learn, and therefore the networks end up learning to predict noise-free building maps which are much more accurate than freely available labels from OpenStreetMap. Data from OpenStreetMap can contain various types of label noise, as both image quality, image annotation tool and labelling effort can all affect fidelity of these observed labels. Data-fusion of high resolution aerial image with satellite images improves performance (Huang et al. 2016; Ji et al. 2019). We use a stricter evaluation regime, where the labels generated by models are evaluated against true noise-free images (obtained after manual labelling) rather than observed labels which may contain noise. A trained model evaluated to have sufficient predictive performance using our evaluation process can be considered a more realistic measure of how well the models in question are able to automatically label images. Mainstream practices measure model performance by comparing against noise observed labels, this process can lead to underestimated model performance, as confirmed by our stricter evaluation process of comparing against ground truth noise-free labels.

The remaining text is organized as follows—Section 2 discusses relevant work on deep learning-based building extraction. Section 3 provides the problem formulation, notations and process flow. Section 4 outlines the deep learning-based segmentation U-Net architecture we have used. Section 5 describes the dataset configurations and label noise injection for simulating registration errors. Section 6 analyzes model prediction confidence and provides arguments for adapting threshold to label noise. Section 7 provides our experimental setup. Section 8 contains our quantitative and qualitative results. Section 9 provides discussion on the results and future work.

## 2. Related work

Currently, deep learning systems are the state-of-the-art for learning automated building extraction from high resolution remotely sensed data (Xu et al. 2018; Yuan 2018; Yang et al. 2018; Chen et al. 2020). Labels for building extraction tasks come from vector shapefiles of individual building polygons. Rasterization of these polygons result in super-

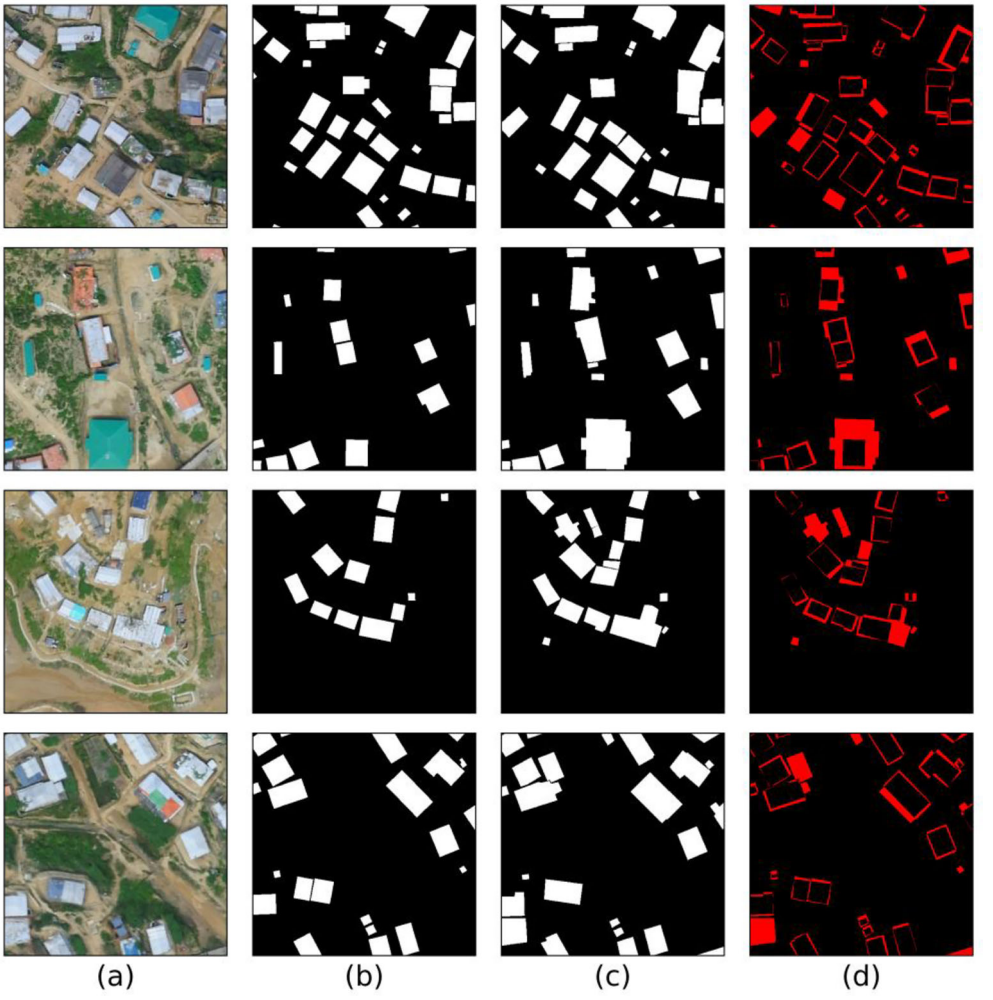



**Figure 1.** Example of label noise in building extraction tasks. The images and labels are from our test set. (a) Image is a 512 × 512 tile from a large ultra-high-resolution aerial image (10 cm resolution) over Kutupalong mega-camp captured by a UAV mounted camera on September 17 2018 (b) Noisy label is extracted from OpenStreetMap vector data (c) True label (ground truth) is obtained after manual relabeling done by the authors (d) The pixels with label noise are located by taking the absolute difference between the true and noisy label matrices, these pixels represent the locations which were either omitted or incorrectly included in the noisy labels with respect to the true labels.

pixel-based labels rather than pixel-based labels i.e. labels for a group connected pixels are assigned rather than labels for individual pixels. Mnih and Hinton (2012) have used the asymmetric Bernoulli noise (ABN) model to deal with omission label noise. It is assumed that there exists a clean distribution which is corrupted and is accessible as the observed labels; a latent variable is used for mapping the two distributions. However, they use high-resolution images of big city like Massachusetts for their task which somewhat alleviates the issue, which is not the case for us. Maas et al. (2016) have employed the noise robust logistic regression model (Bootkrajang and Kabán 2012) for building extraction. In this model, the marginal distribution of the observed labels over every possible state of the clean uncorrupted labels is learned. Maas et al. (2009) have used an adapted random forest classifier where instances are assigned to every possible class with respective

probabilities to utilize outdated building maps contaminated with label noise. It is evident that Maas et al. (2016, 2019) have prioritized the multi-class settings of label noise robust building extraction and focus on non-dense prediction methods.

Zhang et al. (2020) have used a probability transition modular (PTM) after the base model to learn the mappings between the observed and clean distributions. They focus on registration noise (labels are shifted or rotated) in multi-class settings. Mnih and Hinton (2012) have used the translational asymmetric binary noise (TABN) model for the task of handling registration noise which generates different translations for each image patch to learn registration noise free mappings. We have also found that regularization and data augmentation allows reliable learning of noise-free representations from massively noisy labels (Ahmed et al. 2020b). Unavailability of ground truth test labels has been the reason why most studies on deep learning-based building extraction do not even acknowledge or mention label noise in their methodologies. We have observed that the label noise in the Rohingya camp dataset follows the *noisy at random* (NAR) model (Frénay and Verleysen 2014). We confirm this statement by testing our trained models against clean labels which have been relabeled by hand. Virtually all aforementioned works which use a label noise adaptive methodology test against noisy observed labels. They do not test against the ground truth clean labels as obtaining those labels for their large-scale test sets would be very time consuming and laborious Mainstream practices in building extraction test against observed noisy labels, this can reflect misleadingly on the models and wrongly indicate poor performance. That is why opt for using ground truth noise-free labels for testing and gauging true performance. Current literature which compares against noisy and counterpart noise-free labels for evaluation have used a smaller dataset size of 22 million pixels (Ahmed et al. 2020b), while we use a larger test set of 67 million pixels which covers a larger distribution of the data.

## 3. Problem formulation

In the general building extraction case, given satellite/aerial images $\boldsymbol{S} \in \mathbb{R}^{n_h \times n_w \times n_d}$, where $n_h$, $n_w$ and $n_d$ are the height, width and depth (bands/channels) respectively, and corresponding observed label maps $\widetilde{\boldsymbol{M}} \in \{0,1\}^{n_h \times n_w}$, the probability distribution of the label maps given the input images are modelled, where the predicted label maps $\hat{\boldsymbol{M}}$ estimates the mean of the distribution

$$P(\widetilde{\boldsymbol{M}}|\boldsymbol{S}) = \prod_{i=1}^{n_h} \prod_{j=1}^{n_w} P(\widetilde{\boldsymbol{M}}_{i,j}|\boldsymbol{S}) \tag{1}$$

Each $P(\widetilde{\boldsymbol{M}}_{i,j}|\boldsymbol{S})$ is independently and identically distributed. Studies acknowledging label noise assume that the clean uncorrupted binary label maps $\boldsymbol{M} \in \{0,1\}^{n_h \times n_w}$ (which are inaccessible and/or unknown to the learning system) undergo the label noise process and are accessible as the observed labels $\widetilde{\boldsymbol{M}}$. We argue that it is possible to learn the clean noise-free label distribution, $P(\boldsymbol{M}|\boldsymbol{S})$ from its corresponding observed noisy distribution, $P(\widetilde{\boldsymbol{M}}|\boldsymbol{S})$, subject to random label noise. In most label noise robust building extraction studies, the models are trained and tested on data from the same noisy observed label distribution. Therefore, the models are effectively tested against noisy labels. In this work, we acknowledge and address this issue by creating a corresponding clean true label dataset which represents label samples from the true distribution $P(\boldsymbol{M}|\boldsymbol{S})$. We investigate the performance of models *trained solely on noisy observed labels and tested against*

**Table 1.** Notations and descriptions.

| No | Notation | Description |
|---|---|---|
| 1 | $\boldsymbol{S}$ | Satellite/aerial image tensor |
| 2 | $\widetilde{\boldsymbol{M}}$ | Observed noisy dense label tensor |
| 3 | $\boldsymbol{M}$ | True dense label tensor |
| 4 | $\hat{\boldsymbol{M}}$ | Predicted dense label tensor |
| 5 | $\underline{\boldsymbol{S}}^{dataset}$ | Set of all image tensors |
| 6 | $\underline{\widetilde{\boldsymbol{M}}}^{dataset}$ | Set of all observed noisy dense label tensors |
| 7 | $\underline{\boldsymbol{S}}^{train}$ | Set of image tensors selected for training |
| 8 | $\underline{\widetilde{\boldsymbol{M}}}^{train}$ | Set of observed noisy dense label tensors selected for training |
| 9 | $\underline{\boldsymbol{S}}^{test}$ | Set of image tensors selected for testing |
| 10 | $\underline{\widetilde{\boldsymbol{M}}}^{test}$ | Set of observed noisy dense label tensors selected for testing |
| 11 | $\underline{\boldsymbol{M}}^{test}$ | Set of true dense label tensor (de-noised counterpart of $\underline{\widetilde{\boldsymbol{M}}}^{test}$) for testing |
| 12 | $\underline{\hat{\boldsymbol{M}}}^{test}$ | Set of dense label tensors predicted by trained segmentation model |
| 13 | $\psi_{\underline{\boldsymbol{M}}^{test}}(\underline{\widetilde{\boldsymbol{M}}}^{test})$ | Performance metric function considering $\underline{\boldsymbol{M}}^{test}$ as ground truth and $\underline{\widetilde{\boldsymbol{M}}}^{test}$ as labels being evaluated |
| 14 | $\psi_{\underline{\boldsymbol{M}}^{test}}(\underline{\hat{\boldsymbol{M}}}^{test})$ | Performance metric function considering $\underline{\boldsymbol{M}}^{test}$ as ground truth and $\underline{\hat{\boldsymbol{M}}}^{test}$ as labels being evaluated |

*ground-truth clean labels*, and thus delineate the training and testing process. Table 1 summarizes all notation and descriptions for quick lookup.

If we consider a dataset with n images and labels, with the vector of label matrices, $\underline{\widetilde{\boldsymbol{M}}}^{dataset}$ randomly divided into training set $\underline{\widetilde{\boldsymbol{M}}}^{train}$ and testing set $\underline{\widetilde{\boldsymbol{M}}}^{test}$. We first obtain the noise free set of test labels $\underline{\boldsymbol{M}}^{test}$ (de-noised counterpart of $\underline{\widetilde{\boldsymbol{M}}}^{test}$) and labels predicted on the test set $\underline{\hat{\boldsymbol{M}}}^{test}$, by our segmentation model trained on $\underline{\widetilde{\boldsymbol{M}}}^{train}$. Let, $\psi_{\underline{\boldsymbol{M}}^{test}}$ be a performance metric function of the generated labels being evaluated against the clean labels; in most semantic segmentation tasks the precision, recall, Dice coefficient (F1-score) and Jaccard index (IoU) are common performance metrics. Hence, our primary objective is to empirically prove that,

$$\psi_{\underline{\boldsymbol{M}}^{test}}\left(\underline{\hat{\boldsymbol{M}}}^{test}\right) \gg \psi_{\underline{\boldsymbol{M}}^{test}}\left(\underline{\widetilde{\boldsymbol{M}}}^{test}\right) \tag{2}$$

Our proposed evaluation regime is displayed in Figure 2 (region shaded light blue).

## 4. Segmentation model

We used the U-Net model [inspired by the fully convolutional network (FCN) (Long et al. 2015) architecture] as our semantic segmentation model. We use a much smaller network than Ahmed et al. (2020b) (1.94 million parameters versus approximately 31 million parameters). The encoding blocks ($D_1$, $D_2$, $D_3$, $D_4$) learn higher order features for classification of pixels, shrinking the row and column size while increasing depth, whereas the decoding blocks ($U_1$, $U_2$, $U_3$, $U_4$) incorporate structure integrity back into prediction. This network architecture is appropriate for semantic segmentation as it prioritizes both classification and prediction structure; furthermore, prediction density is maximum, as row and column size of predictions are equal to those of the input image (Figure 3). We used variable dropout rates, to improve generalization. D1 and U4 have dropout rates of 0.1, D2 and U3 also have dropout rates of 0.1, D3 and U2 have dropout rates of 0.2, and D4 and U1 have dropout rates of 0.3. Dropout layers have been placed between the first and second convolutional layers of each block.

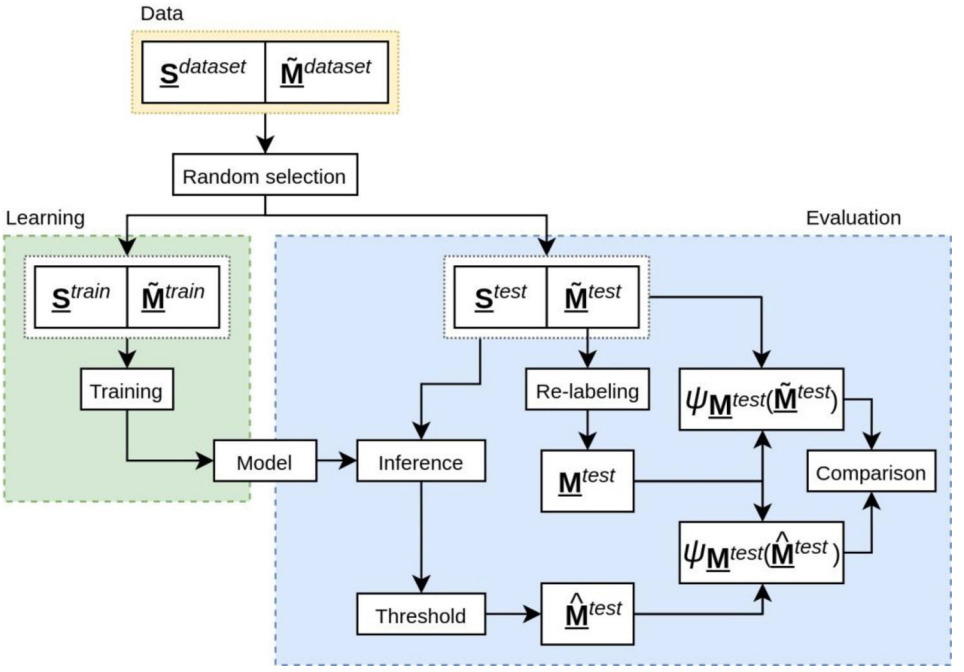



**Figure 2.** Process flow of learning and proposed evaluation regime utilized in this work.

We used the exponential linear unit (ELU) as the activation function for the hidden layers and use the sigmoid activation function for the final pixel level predictions in the last layer. As our task involves semantic binary segmentation, we use the binary cross-entropy loss (Eq. (3)), and optimize this function using the Adam optimizer (Kingma and Ba 2014) to learn the parameters of the network.

$$-\frac{1}{n_h \times n_w}\sum_{i=1}^{n_h}\sum_{j=1}^{n_h}\left[\widetilde{\boldsymbol{M}}_{i,j} log\hat{\boldsymbol{M}}_{i,j} + \ (1-\widetilde{\boldsymbol{M}}_{i,j}) log(1-\hat{\boldsymbol{M}}_{i,j})\right] \tag{3}$$

## 5. Data

The United Nations International Organization for Migration (IOM) collected ultra-high-resolution aerial images of Rohingya camps using camera mounted unmanned aerial vehicles (UAVs). We have access to ultra-high-resolution images of Rohingya refugee camps capturing fine grained details (Figure 1). Data catalog of all collected images is available.[1] These large raster images were created by stitching together images captured by the UAV and are available as large raw raster files. Image of Kutupalong, the largest of the refugee camps, captured on September 17 2018, is used (Figure 4(a)). The raster image has a spatial resolution of 10 cm (0.1 m). The labels are obtained from OpenStreetMap vector data which contain shapefiles of individual building footprint polygons (Figure 4(b)). OpenStreetMap data have been observed to contain label noise, especially for rapidly changing semi-built environments such as the Rohingya camps. Virtually all camp buildings are single storied structures with corrugated sheet metal roofs. Most of these buildings are cubic and utilitarian and therefore the roofs of the buildings in the images approximate the building footprints. The building roofs cover only the building

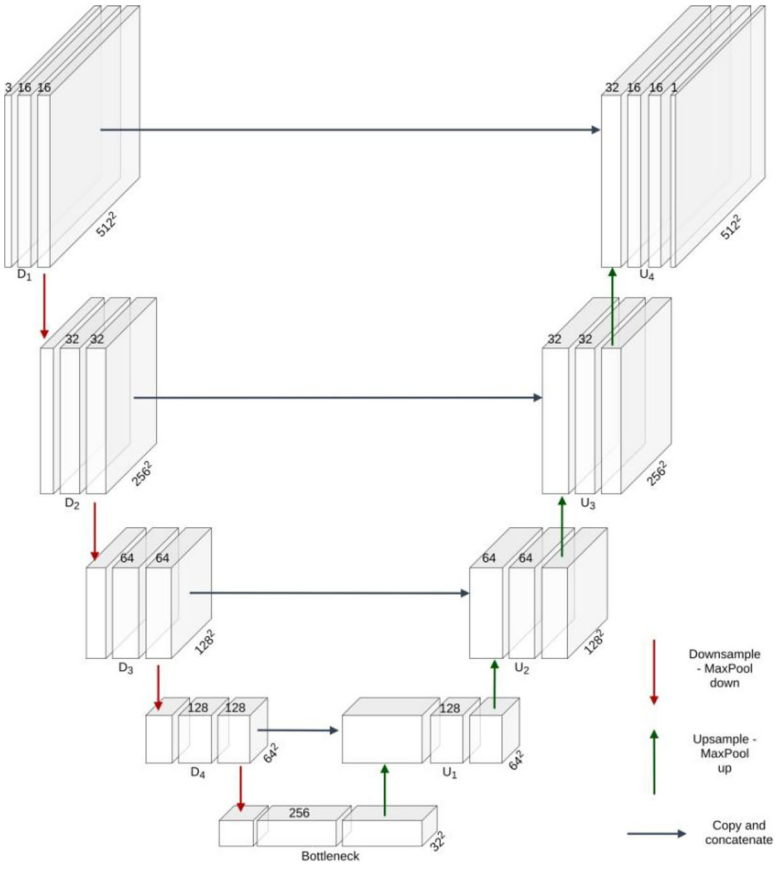



**Figure 3.** Network architecture of U-Net used as segmentation model.

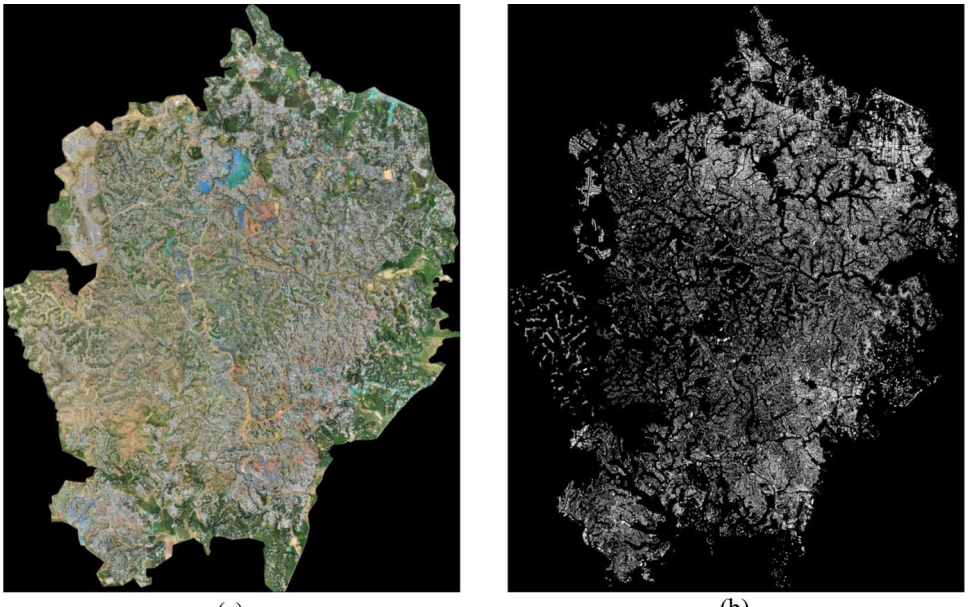



**Figure 4.** Image and rasterized labels. (a) Ultra-high-resolution aerial image of Kutupalong camp (b) Observed labels from OpenStreetMap containing approximately 130,000 building polygons.

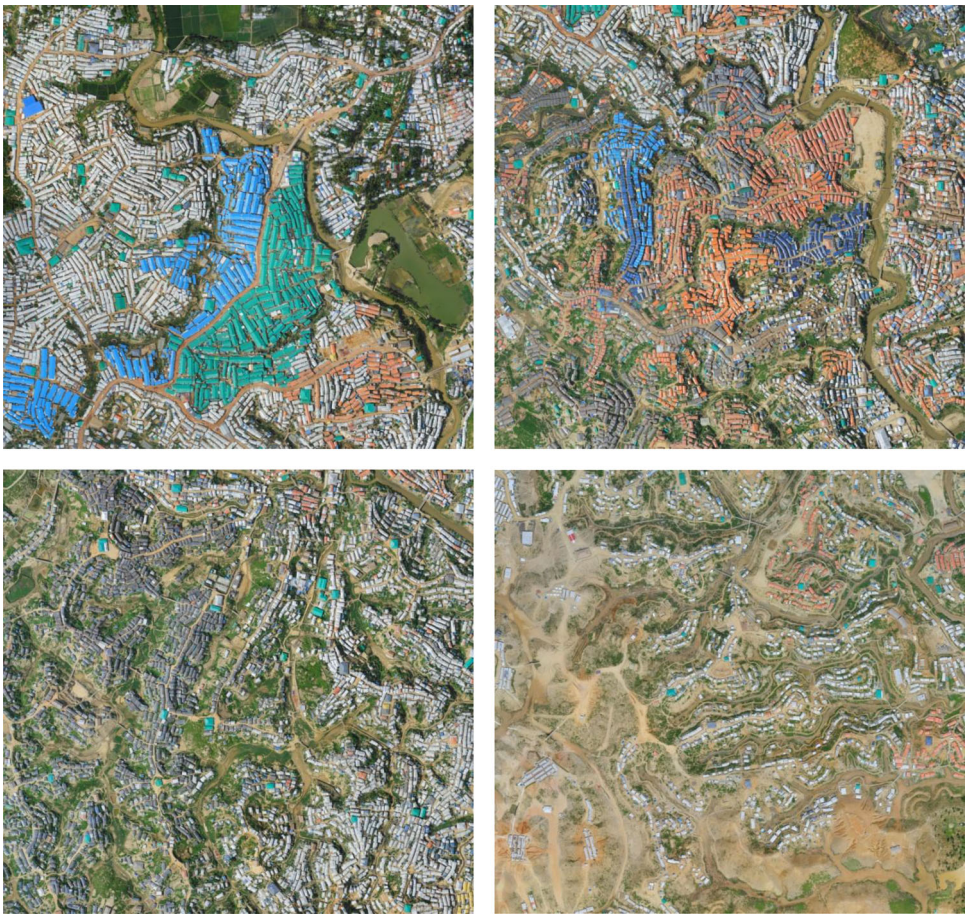


**Figure 5.** Variance in colour, shape, size and density of refugee camp buildings.

rooftop as the camp buildings are densely packed with focus on utility rather than aesthetics/design. Owing to the simplicity of the camp building design, the rooftops visible in the aerial image mosaic closely resemble the actual building footprints. We have used the administrative camp boundary shapefile for Kutupalong to clip the raster image and vector labels. The labels from OpenStreetMap and the aerial image mosaics from UN IOM both follow the WGS 84 coordinate reference system (CRS), and did not require any further correction. We then rasterized the vector labels with respect to the image's metadata i.e. scale, spatial resolution, coordinate reference system, etc. Pixels falling within building polygons are assigned to the positive class and the remaining are assigned to the negative class. Zoomed in views of image in Figure 4(a) are shown in Figure 5.

The raster image and label are cropped to $512 \times 512$ sized tiles. Empty tiles without buildings are removed, leading to 5160 image tiles. 50% of the image tiles are selected as training data i.e. 2580 tiles (approximately 680 million pixels) while the rest are used for testing and validation. We refer to this training set as TRAIN-NOISY-EMP. 10% of the remaining image tiles i.e. 258 tiles (approximately 67 million pixels) are randomly selected for manual relabeling to obtain ground truth clean labels. We refer to this clean test set as TEST-CLEAN. Manual labelling of the test set of 258 large $512 \times 512$ image tiles has taken approximately 62 h, which averages to approximately 4.2 image tiles labelled per

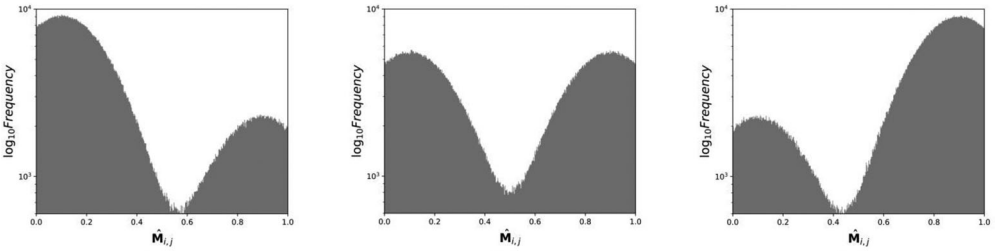



**Figure 6.** Simulated soft label predictions. Each bimodal distribution is approximated from $2 \times 10^6$ samples. The distribution of negative samples $\widetilde{N}(0.1, 0.2)$ and the distribution of positive samples $\widetilde{N}(0.9, 0.2)$. (a) Negative heavy imbalanced dataset, 80% negative and 20% positive samples, (b) Balanced dataset, 50% negative and 50% positive samples, (c) Positive heavy imbalanced dataset, 20% negative and 80% positive samples.

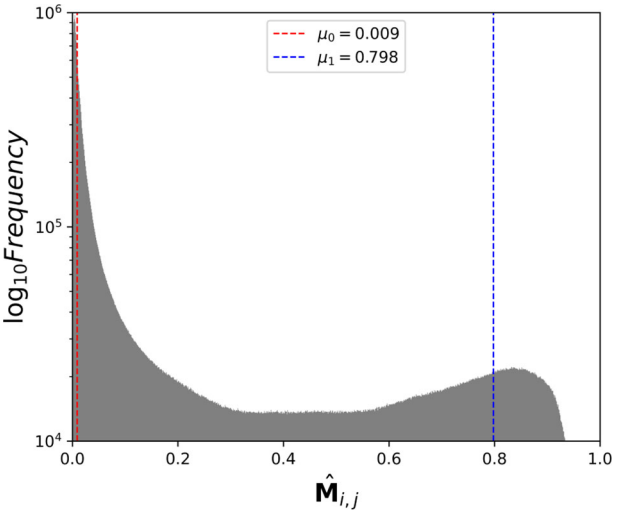



**Figure 7.** Distribution of pixel-wise predictions on randomly selected subset of training data.

hour. The corresponding test labels with label noise (from OpenStreetMap vector data) are referred to as TEST-NOISY-EMP.

## 6. Label noise adaptive prediction threshold

Dense segmentation models provide probability values between 0 and 1. Prior to performance evaluation, the soft labels are converted to hard labels using a threshold value. The most commonly used threshold value of 0.5 assumes that the model performance is balanced on both classes (Figure 6(b)). However, in the presence of label noise in an already imbalanced dataset, such assumptions do not hold (Figure 6(a,b)). We have observed that the primary source of label noise in predictions occurs because model confidence reduces around building edges, leading to under-detection. This is because the majority of noisy pixel labels in the training set are also situated around building edges. We analyze the distribution of pixel-wise soft label predictions to identify suitable threshold values for our case. As Figure 7 shows our case is negative heavy imbalanced dataset.

We segregated pixel-wise predictions based on class label (positive/building or negative/non-building) and confidence (confident or unconfident). We approximated the

distribution of pixel-wise predictions on TRAIN-NOISY-EMP using a four component Gaussian mixture model. The components are as follows –

i. distribution of confident negative predictions,

$$P\left(0 \leq \hat{\boldsymbol{M}}_{i,j} \leq \tau_0\right) \sim N\left(\mu_0, \sigma_0^2\right); \;\; \tau_0 \ll 0.5 \tag{4}$$

ii. distribution of confident positive predictions,

$$P\left(\tau_1 \leq \hat{\boldsymbol{M}}_{i,j} \leq 1\right) \sim N\left(\mu_1, \sigma_1^2\right); \tau_1 \gg 0.5, \tag{5}$$

iii. distribution of unconfident negative predictions,

$$P\left(\tau_0 < \hat{\boldsymbol{M}}_{i,j} < 0.5\right) \sim N(\mu_{u^-}, \sigma_{u^-}^2), \tag{6}$$

iv. distribution of unconfident positive predictions,

$$P\left(0.5 < \hat{\boldsymbol{M}}_{i,j} < \tau_1\right) \sim N(\mu_{u^+}, \sigma_{u^+}^2), \tag{7}$$

We randomly selected 5% of the training data for modelling the Gaussian mixture. We are interested in $\mu_0$ and $\mu_1$, calculated from the components of the Gaussian mixture (Figure 7). It is evident from Figure 7 that the class imbalance skews the distribution to the negative predictions and threshold value plays an important role in generating hard labels. The calculated class priors $P(\hat{\boldsymbol{M}}_{i,j} = 0) \approx 0.7684$ and $P(\hat{\boldsymbol{M}}_{i,j} = 1) \approx 0.2316$ (at the general threshold of 0.5) also confirm imbalance.

We can confirm that the optimal threshold value $\tau$ for our case should be much closer to $\mu_0$ than $\mu_1$. Eq. (8) shows how threshold value determines the final predicted hard labels. We varied the threshold values from the set $\tau = \{0.3,\ 0.4,\ 0.5,\ 0.6,\ 0.7\}$ in Eq. (8), to see its effects on independent performance evaluation.

$$\hat{\boldsymbol{M}}_{i,j} = \begin{cases} 1, & if \hat{\boldsymbol{M}}_{i,j} > \tau \\ 0, & if \hat{\boldsymbol{M}}_{i,j} \leq \tau \end{cases} \tag{8}$$

## 7. Experimental setup

To test even more challenging settings, we introduced synthetic label noise in both training and testing data. Labels are rotated by 15 degrees about the centre to simulate massive registration noise. Our training set with added synthetic noise is referred to as TRAIN-NOISY-EMP-SYN. The empirically noisy test set with added synthetic noise is referred to as TEST-NOISY-EMP-SYN, clean test set with added synthetic noise is referred to as TEST-NOISY-SYN. Figure 8 shows the effects of synthetic label noise introduced through rotated labels of the training set. Figure 1(b) contains sample images from TEST-NOISY-EMP and Figure 1(c) contains the corresponding sample images from TEST-CLEAN. Table 2 shows all datasets derived for training and testing. We trained two separate segmentation models (on TRAIN-NOISY-EMP and on TRAIN-NOISY-EMP-SYN). Each trained model is evaluated against the 5 test data configurations. As mentioned previously, we are particularly interested in the cases where the model is trained on TRAIN-NOISY-EMP and tested against TEST-CLEAN. The deep segmentation networks have been implemented in TensorFlow 2.0 using its corresponding Python 3 API and trained on a Google Colab VM. The models are trained for 5 epochs using a step size of 5 and a batch size of 16, and take approximately 40 min to train on the GPU enabled backend

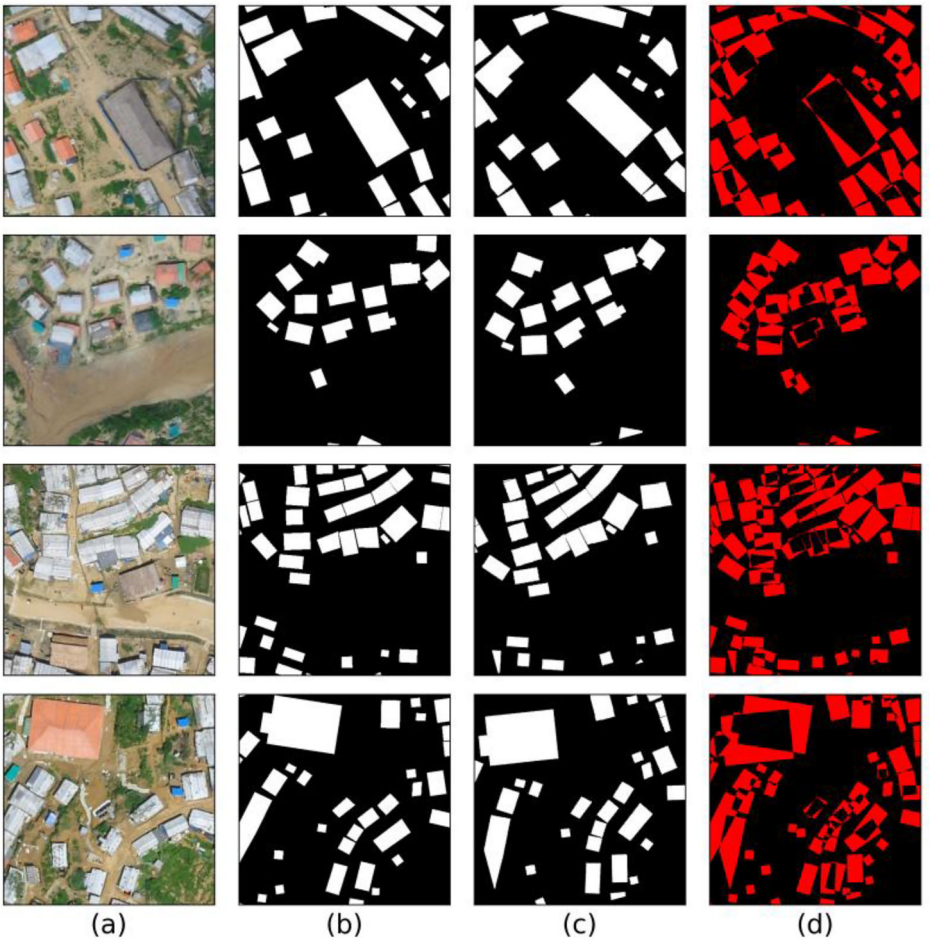



Figure 8. Effects of synthetic label noise, (a) Image (b) TRAIN-NOISY-EMP (c) TRAIN-NOISY-EMP-SYN (d) Pixels with label noise due to only synthetic noise.

configuration for the Google Colab environment. We believe that loss itself is an indicator of performance against observed (and potentially noisy labels), that is why we use a stricter evaluation regime where performance is measured against the true labels.

## 8. Results

### 8.1. Quantitative evaluation

The total number of true positives, true negatives, false positives and false negatives for the 67 million pixels of each test set are calculated, which are then used for calculating performance metrics for semantic binary segmentation with class imbalance. We calculated the precision, recall, F1-score (also known as Dice score) and IoU (also known as Jaccard index). With two training sets and 4 test sets, each evaluated at 5 threshold values, we analyze 40 configurations (Table 3). A general trend can be observed in terms of using training on labels with only empirical noise versus training on labels with both empirical and synthetic noise. All configurations trained on TRAIN-NOISY-EMP-SYN have poorer

**Table 2.** The different data configurations used for training and testing.

| No | Data | Remarks |
|---|---|---|
| 1 | TRAIN-NOISY-EMP | Training labels containing empirical label noise |
| 2 | TRAIN-NOISY-EMP-SYN | Training labels containing both empirical and synthetic label noise |
| 3 | TEST-CLEAN | Ground truth test labels obtained by manual relabeling |
| 4 | TEST-NOISY-EMP | Test labels containing empirical label noise |
| 5 | TEST-NOISY-SYN | Test labels containing synthetic label noise |
| 6 | TEST-NOISY-EMP-SYN | Test labels with containing both empirical and synthetic label noise |

**Table 3.** Performance metrics of evaluated configurations.

| No | Training data | Testing data | $\tau$ | Precision | Recall | F1-score | IoU |
|---|---|---|---|---|---|---|---|
| 1 | TRAIN-NOISY-EMP | TEST-CLEAN | 0.3 | 0.87548 | 0.85309 | 0.86414 | 0.76078 |
| 2 | | | 0.4 | 0.92018 | 0.77551 | 0.84168 | 0.72663 |
| 3 | | | 0.5 | 0.95045 | 0.67742 | 0.79104 | 0.65431 |
| 4 | | | 0.6 | 0.96776 | 0.56231 | 0.71132 | 0.55197 |
| 5 | | | 0.7 | 0.9787 | 0.41937 | 0.58715 | 0.41558 |
| 6 | | TEST-NOISY-EMP | 0.3 | 0.60627 | 0.807 | 0.69238 | 0.5295 |
| 7 | | | 0.4 | 0.64975 | 0.74803 | 0.69544 | 0.53308 |
| 8 | | | 0.5 | 0.69284 | 0.67456 | 0.68357 | 0.51927 |
| 9 | | | 0.6 | 0.73568 | 0.58392 | 0.65107 | 0.48266 |
| 10 | | | 0.7 | 0.78381 | 0.45879 | 0.57879 | 0.40726 |
| 11 | | TEST-NOISY-SYN | 0.3 | 0.51673 | 0.53799 | 0.52715 | 0.35791 |
| 12 | | | 0.4 | 0.52602 | 0.47368 | 0.49848 | 0.33198 |
| 13 | | | 0.5 | 0.53584 | 0.40807 | 0.4633 | 0.30149 |
| 14 | | | 0.6 | 0.54562 | 0.33874 | 0.41798 | 0.26421 |
| 15 | | | 0.7 | 0.55874 | 0.25581 | 0.35095 | 0.21282 |
| 16 | | TEST-NOISY-EMP-SYN | 0.3 | 0.37167 | 0.52727 | 0.43601 | 0.27878 |
| 17 | | | 0.4 | 0.37879 | 0.46477 | 0.4174 | 0.26374 |
| 18 | | | 0.5 | 0.38631 | 0.40086 | 0.39345 | 0.24491 |
| 19 | | | 0.6 | 0.39511 | 0.33424 | 0.36213 | 0.2211 |
| 20 | | | 0.7 | 0.40872 | 0.25498 | 0.31404 | 0.18627 |
| 21 | TRAIN-NOISY-EMP-SYN | TEST-CLEAN | 0.3 | 0.63636 | 0.55544 | 0.59315 | 0.42162 |
| 22 | | | 0.4 | 0.69319 | 0.3163 | 0.43439 | 0.27746 |
| 23 | | | 0.5 | 0.69773 | 0.12819 | 0.21658 | 0.12144 |
| 24 | | | 0.6 | 0.72978 | 0.02835 | 0.05457 | 0.02805 |
| 25 | | | 0.7 | 0.89665 | 0.0019 | 0.0038 | 0.0019 |
| 26 | | TEST-NOISY-EMP | 0.3 | 0.4569 | 0.54477 | 0.49698 | 0.33066 |
| 27 | | | 0.4 | 0.50971 | 0.3177 | 0.39143 | 0.24334 |
| 28 | | | 0.5 | 0.51329 | 0.12882 | 0.20595 | 0.1148 |
| 29 | | | 0.6 | 0.52595 | 0.02791 | 0.053 | 0.02722 |
| 30 | | | 0.7 | 0.42601 | 0.00124 | 0.00246 | 0.00123 |
| 31 | | TEST-NOISY-SYN | 0.3 | 0.67285 | 0.62751 | 0.64939 | 0.48081 |
| 32 | | | 0.4 | 0.7552 | 0.36819 | 0.49503 | 0.32893 |
| 33 | | | 0.5 | 0.82947 | 0.16283 | 0.27222 | 0.15755 |
| 34 | | | 0.6 | 0.90438 | 0.03753 | 0.07207 | 0.03738 |
| 35 | | | 0.7 | 0.967 | 0.00219 | 0.00438 | 0.00219 |
| 36 | | TEST-NOISY-EMP-SYN | 0.3 | 0.48745 | 0.61943 | 0.54557 | 0.37511 |
| 37 | | | 0.4 | 0.56442 | 0.37495 | 0.45058 | 0.2908 |
| 38 | | | 0.5 | 0.63104 | 0.16879 | 0.26634 | 0.15363 |
| 39 | | | 0.6 | 0.67987 | 0.03845 | 0.07278 | 0.03776 |
| 40 | | | 0.7 | 0.49411 | 0.00153 | 0.00305 | 0.00153 |

performance with respect to the corresponding performance of configurations trained on TRAIN-NOISY-EMP. Test labels also showed general trends, testing against clean labels (TEST-CLEAN) showed higher performance with respect to testing against noisy labels (TEST-NOISY-EMP, TEST-NOISY-SYN and TEST-NOISY-EMP-SYN). Performance against noisy labels depended on noise level and type. Higher noise levels in test labels resulted in higher underestimation of model performance. Empirical noisy test labels from OpenStreetMap (TEST-NOISY-EMP) are more accurate than our synthetic noisy test labels (TEST-NOISY-EMP), as indicated by performance against these labels.

**Table 4.** Performance metrics of evaluated fidelity of label sets.

| Evaluated labels | Evaluated against | Precision | Recall | F1-score | IoU |
|---|---|---|---|---|---|
| TEST-NOISY-EMP | TEST-CLEAN | 0.8183 | 0.59904 | 0.69171 | 0.52871 |
| TEST-NOISY-SYN | | 0.55426 | 0.51874 | 0.53591 | 0.36603 |
| TEST-NOISY-EMP-SYN | | 0.5358 | 0.36803 | 0.43634 | 0.27905 |

Introduction of label noise in the form of label rotation by just 15 degrees in the training set can completely offset model performance, yielding very poor performing configurations. In our case where the performance bottleneck lies within increasing detection rate, recall is a better indicator of performance, and using threshold values closer to $\mu_0$ improved detection rate and recall. The distribution of confident negative predictions (Eq. (4)) contained mostly non-building areas and the distribution of confident positive predictions (Eq. (5)) contains the building pixels at the center/middle of superpixels. The unconfident predictions (negative (Eq. (6)) and positive (Eq. (7)) arise from border of building superpixels and regions between building superpixels. Using a non-uniform threshold value alleviates the issue providing more accurate and less noisy dense labels (Table 3).

Threshold values also play an important role, as Figure 7. shows, a general threshold value of 0.5 might not be suitable for our case due to imbalanced classes. The general pattern of lower thresholds yielding lower precision and higher recall holds for our case. However, we observed that lower threshold values increase both F1-score and IoU, due to class imbalance. This indicates that a major source of decrease in performance in terms of F1-score and IoU can be attributed to undetected building pixels (false negatives). We considered the configuration in the first row of Table 3 to be our best performing model, and refer to its performance on against our true ground truth labels as $\psi_{\underline{\mathbf{M}}^{test}}^{F1-score}\left(\hat{\underline{\boldsymbol{M}}}^{test}\right) = 0.86414$ and $\psi_{\underline{\mathbf{M}}^{test}}^{IoU}\left(\hat{\underline{\boldsymbol{M}}}^{test}\right) = 0.76078$. We then evaluate the fidelity of our noisy test labels by comparing against the ground truth clean test labels. The noisy labels can be thought of as outputs a hypothetical segmentation model producing noisy predictions. We reconfirmed that the fidelity of labels in increasing order is TEST-NOISY-EMP-SYN, TEST-NOISY-SYN and TEST-NOISY-EMP. The first row in Table 4 is of special interest as it records the fidelity of labels from freely available and commonly used OpenStreetMap vector data (TEST-NOISY-EMP). We consider the performance/fidelity of the observed empirical noisy labels as $\psi_{\underline{\mathbf{M}}^{test}}^{F1-score}\left(\widetilde{\underline{\boldsymbol{M}}}^{test}\right) = 0.69171$ and $\psi_{\underline{\mathbf{M}}^{test}}^{IoU}\left(\widetilde{\underline{\boldsymbol{M}}}^{test}\right) = 0.52871$.

### 8.2. Qualitative evaluation

Aside from using numeric performance metrics for model evaluation, we also performed qualitative assessment of labels predicted by different configurations. Figure 9 shows some examples of model predictions on independent test images. Using lower threshold values increases detection rate of buildings, but densely packed buildings may lead to 'leaky' labels as the area between two closely situated buildings may also be predicted as buildings (Figure 9(d)). Higher threshold values should be used when 'seperability' of individual building superpixels is favoured over detection rate, as it can lead to loss in structural integrity around edges/boundaries of building superpixels (Figure 9(f)). As Figure 9(g) shows, test labels generated by model trained against labels with empirical and synthetic

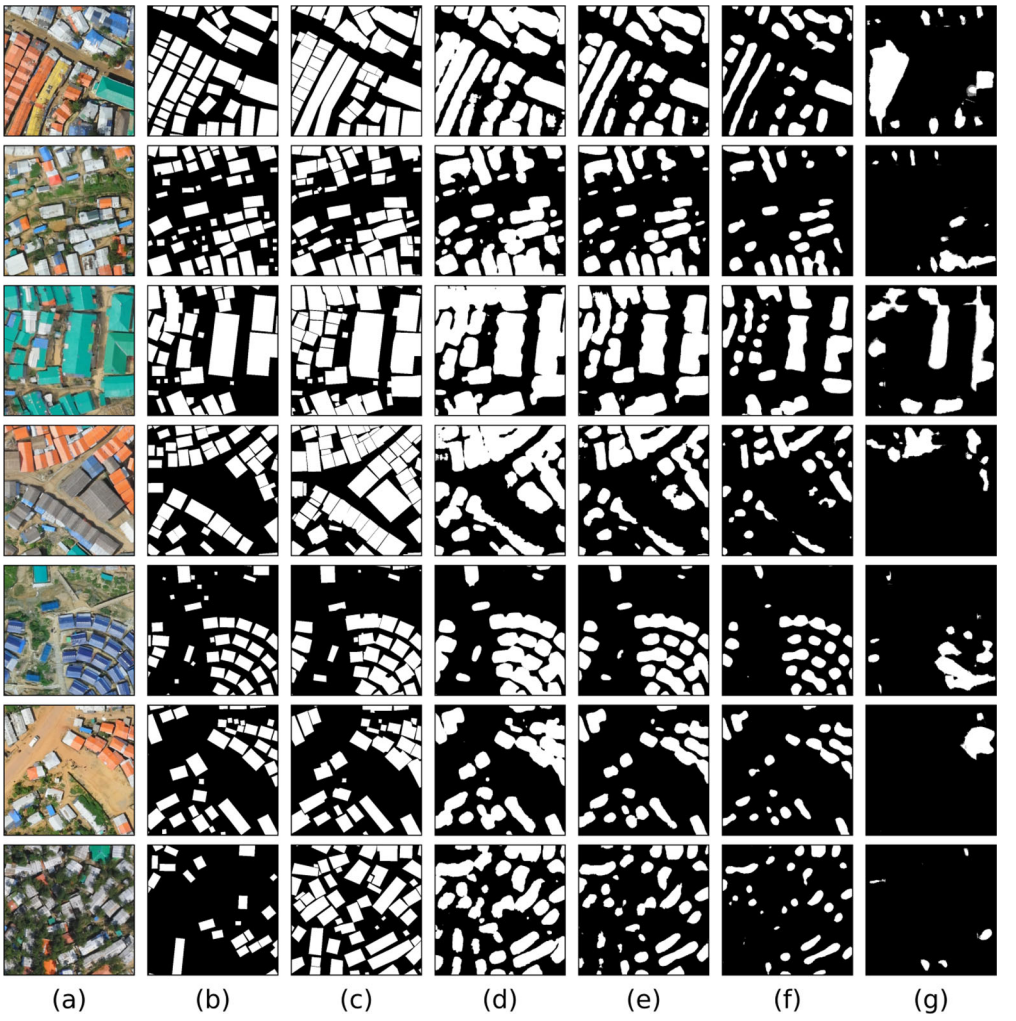



**Figure 9.** Some examples of predicted test labels. (a) Images (b) Observed noisy labels (c) Ground truth clean labels (d) Predictions from model trained on TRAIN-NOISY-EMP with $\tau = 0.3$ (e) Predictions from model trained on TRAIN-NOISY-EMP with $\tau = 0.5$ (f) Predictions from model trained on TRAIN-NOISY-EMP with $\tau = 0.7$ (g) Predictions from model trained on TRAIN-NOISY-EMP-SYN with $\tau = 0.3$.

label noise yielded unusable predictions, with very low detection rate and very high false negative rate.

The outputs of hidden blocks of a trained model are shown in Figure 10. As height and width of tensors are reduced as they go towards the bottleneck, building position activations get stronger while the structural integrity reduces ($D_1$ – $D_4$ in Figure 10). The skip connections from downsampling blocks re-incorporate structure into activations ($U_1$ – $U_4$ in Figure 10) to yield final dense predictions which have high levels of structure as well as discriminative representations. As Figure 10(b) shows, model trained on TRAIN-NOISY-EMP-SYN fail to learn useful representations as introducing synthetic registration errors greatly reduces spatial correspondence between images and noisy labels.

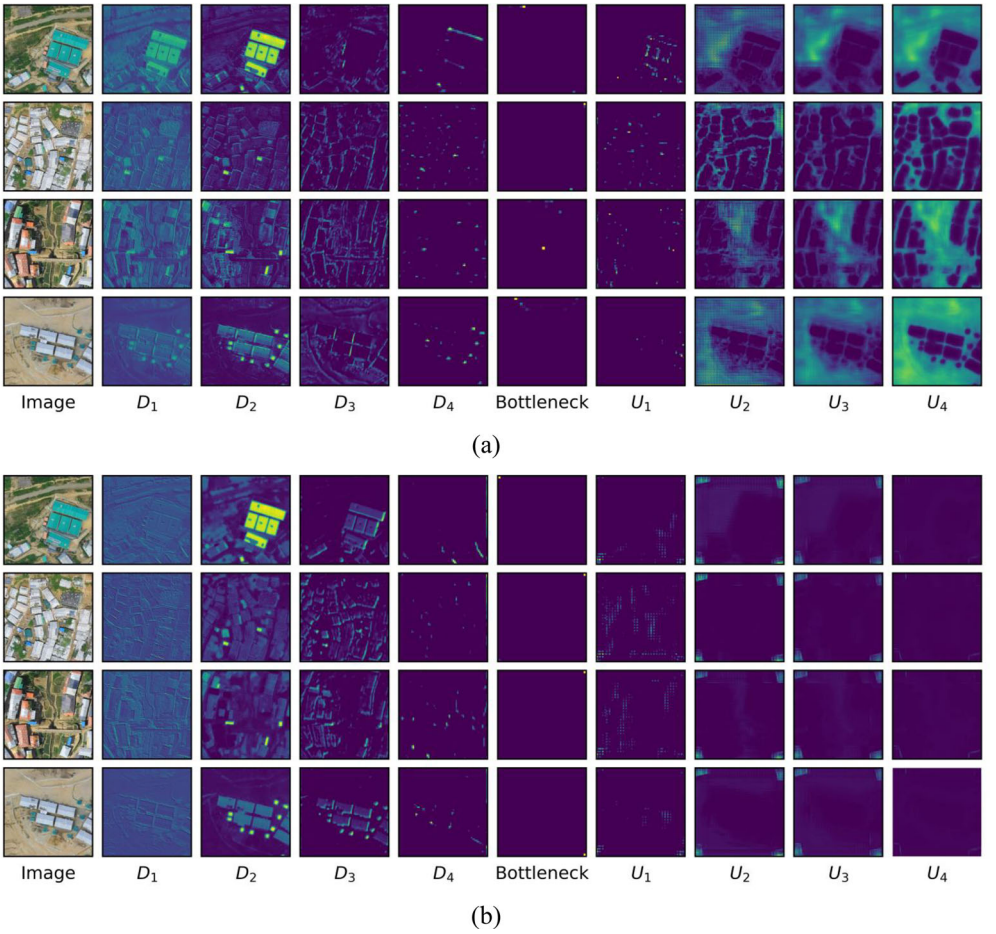



**Figure 10.** Examples of outputs of internal blocks of trained segmentation models. (a) segmentation model trained using TRAIN-NOISY-EMP, (b) segmentation model trained using TRAIN-NOISY-EMP-SYN.

## 9. Discussion

Considering the quantitative analysis of results from Section 8.1, we can confirm our hypothesis in Eq. (2), as shown by the results from Tables 3 and 4. Both performance metrics, F1-score and IoU, are higher than those of commonly used noisy labels form OpenStreetMap i.e. $\psi_{\underline{\boldsymbol{M}}^{test}}^{F1-score}\left(\hat{\underline{\boldsymbol{M}}}^{test}\right) \gg \psi_{\underline{\boldsymbol{M}}^{test}}^{F1-score}\left(\widetilde{\underline{\boldsymbol{M}}}^{test}\right)$ and $\psi_{\underline{\boldsymbol{M}}^{test}}^{IoU}\left(\hat{\underline{\boldsymbol{M}}}^{test}\right) \gg \psi_{\underline{\boldsymbol{M}}^{test}}^{IoU}\left(\widetilde{\underline{\boldsymbol{M}}}^{test}\right)$. Our best performing model generates labels which improve F1-score by 17.2% and improve IoU by 23.2%. This staggering difference can be attributed to label noise in commonly used labels, and confirms that deep learning-based segmentation models can inherently learn intrinsic noise-free representations from noisy labels. Our more strict evaluation method is obviously more realistic and has rarely been adopted for building extraction tasks. Our methodology starkly differs from existing works in this matter. Our models are trained on only observed noisy labels and tested against actual ground truth clean labels that we have annotated carefully. This evaluation method also shows the paradoxical behaviour of deep segmentation networks in terms of noise-robustness, as models trained on noisy data have poor performance when tested using noisy observed labels and have substantial performance improvement when tested using noise-free ground truth

labels. Our stricter evaluation method is obviously more realistic and has rarely been explored in empirical settings. Furthermore, we provide empirical evidence that even a relatively small vanilla U-Net (Ronneberger et al. 2015) configured with only 1.94 million parameters can learn to produce significantly less noisy and more accurate labels (compared to the observed noisy distribution it has learned from) during independent testing.

We show that case specific statistical analysis of pixel-wise soft label predictions can reveal noise adaptive threshold values used for obtaining hard labels. Simulated registration errors can completely overthrow model learning, expressed as poor performance on independent test sets. It is showed that, adapting the threshold value to label noise in predictions arising from false negatives, can improve overall performance. It is also important to note that threshold value has no effect on training regime as soft labels are used for backpropagation; it is only during inference when predictions need to be converted to hard labels using the threshold value. Evaluation of a properly trained model against noisy labels can underestimate performance, we argue that label noise robust segmentation models should be tested against ground truth labels rather than noisy labels from the distribution used for training. Labeling evenly a small randomly selected subset of test labels can provide important insight about label fidelity and model performance.

Since most camp buildings are single storied structures; the roofs of buildings are often obstructed by trees and vegetation (bottom rows of Figure 9). Image inpainting based methods can be used to improve detection of these buildings. When the assumption that no label noise is present in a scenario does not hold, unsupervised and semi-supervised methods can also be explored in lieu of a fully supervised learning routine. Ultra-high-resolution aerial images and dense labels are very memory intensive in terms of both storage and processing. Future studies can focus on optimizing data storage and processing of large-scale memory intensive datasets. Scale robust models can be developed by training and testing against image/label tiles with varying spatial resolutions, to ensure compliance with newly collected aerial images having different spatial resolutions. There is also room for using high resolution aerial images for learning image super-resolution, for times when remotely sensed images of such high resolution are not available. Crowdsourcing can be used for remote sensing and deep learning-based applications (Saralioglu and Gungor 2021) and can play an important role in refugee management as well, because crowdsourcing can be used to obtain noise free labels relatively quickly by employing a larger number of humans in the labeling process.

Our contributions are as follows

- We have developed deep learning-based segmentation models which can produce ultra-high-resolution building maps of Rohingya refugee camps at 10 cm spatial resolution.
- Based on statistical analysis of model prediction confidence, we propose a stricter and more realistic evaluation routine for interpreting model performance in conjunction with label noise.
- More challenging datasets have been generated by injecting synthetic label noise in the form of simulated registration errors
- The effects of real world and simulated label noise on building extraction tasks have been analyzed.
- We re-label 67 million pixels of testing data by hand to obtain clean dense labels and then estimate fidelity using our proposed evaluation routine.
- When tested against our carefully annotated clean labels, we discover that using noise adaptive threshold yields a better performing semantic segmentation model.

- We have made our rigorously evaluated models and code implementations publicly available at https://github.com/nahian-ahmed/bulding-extraction. Our processed dataset (including ultra-high-resolution aerial images of Rohingya camps, labels with simulated and empirical noise, and our relabeled clean labels) has been made available at https://drive.google.com/file/d/19WClVDl4sMhtL3lY_2EfhevwkqjNX69S. Please refer to Section 5 for link to data catalog of all aerial images collected by United Nations IOM.

## Notes

1. Data catalog of all aerial images collected by UN IOM is available at https://data.humdata.org/dataset/e0f0a2ca-9e9e-4fdd-a214-9effb2e3d083/resource/293e01c5-d53e-4db1-a114-c8e646523350/download/iom-npm-uav-repository.pdf.

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## Disclosure statement

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